QUASI-SEPARATEDNESS AND SEPARATEDNESS DESCEND ALONG A SURJECTIVE UNIVERSALLY CLOSED MORPHISM OF ALGEBRAIC SPACES

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof uses two lemmas of the same chapter that are counted separately in this lane; they are consumed as inputs inside the counted proof span and are declared as navigation pointers for the assembly.

Editorial note. The five references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- A universally closed morphism of algebraic spaces is quasi-compact. Lemma lemma-universally-closed-quasi-compact, used at proof lines 1806-1807. Makes the right vertical arrow of the diagonal square quasi-compact.
- The diagonal is representable, locally of finite type and a monomorphism. Lemma lemma-properties-diagonal, used at proof lines 1823-1825. Supplies the three properties needed to turn universal closedness into a closed immersion.
- A universally closed representable morphism is a closed immersion. Etale Morphisms, Lemma etale-lemma-characterize-closed-immersion, with Spaces, Lemma spaces-lemma-representable-transformations-property-implication, used at proof lines 1826-1831. Delivers the closed immersion conclusion for Delta_{Y/B}.
- Separatedness of an algebraic space over S is separatedness relative to Spec(Z). Properties of Spaces, Definition spaces-properties-definition-separated, used at proof lines 1791-1794. Reduces parts (1) and (2) to parts (3) and (4).
- Universally closed and surjective morphisms interact with quasi-compactness of base change. used at proof lines 1802-1815. Is the mechanism of the descent of quasi-separatedness.

Source reference. Original lemma 04XV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6408–6408. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03HK: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4570–4570. Navigation pointer into the Stacks

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Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 04XW: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6409–6409. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02YO: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3883–3883. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original definition 03BS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4359–4359. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let B be an algebraic space over S. Let $f : X \to Y$ be a surjective universally closed morphism of algebraic spaces over B.*

(1) *If X is quasi-separated, then Y is quasi-separated.*
(2) *If X is separated, then Y is separated.*
(3) *If X is quasi-separated over B, then Y is quasi-separated over B.*
(4) *If X is separated over B, then Y is separated over B.*

Proof. Parts (1) and (2) are a consequence of (3) and (4) for $S = B = \mathrm{Spec}(\mathbf{Z})$ (see Properties of Spaces, Definition 03BS). Consider the commutative diagram

$$\begin{array}{ccc} X & \xrightarrow{\Delta_{X/B}} & X \times_B X \\ \downarrow & & \downarrow \\ Y & \xrightarrow{\Delta_{Y/B}} & Y \times_B Y \end{array}$$

The left vertical arrow is surjective (i.e., universally surjective). The right vertical arrow is universally closed as a composition of the universally closed morphisms $X \times_B X \to X \times_B Y \to Y \times_B Y$. Hence it is also quasi-compact, see Lemma 04XW.

Assume X is quasi-separated over B, i.e., $\Delta_{X/B}$ is quasi-compact. Then if Z is quasi-compact and $Z \to Y \times_B Y$ is a morphism, then $Z \times_{Y \times_B Y} X \to Z \times_{Y \times_B Y} Y$ is surjective and $Z \times_{Y \times_B Y} X$ is quasi-compact by our remarks above. We conclude that $\Delta_{Y/B}$ is quasi-compact, i.e., Y is quasi-separated over B.

Assume X is separated over B, i.e., $\Delta_{X/B}$ is a closed immersion. Then if Z is affine, and $Z \to Y \times_B Y$ is a morphism, then $Z \times_{Y \times_B Y} X \to Z \times_{Y \times_B Y} Y$ is surjective and $Z \times_{Y \times_B Y} X \to Z$ is universally closed by our remarks above. We conclude that $\Delta_{Y/B}$ is universally closed. It follows that $\Delta_{Y/B}$ is representable, locally of finite type, a monomorphism (see Lemma 03HK) and universally closed, hence a closed immersion, see Étale Morphisms, Lemma 04XV (and also the abstract principle Spaces, Lemma 02YO). Thus Y is separated over B. □